

1. Suppose registers f3 and f4 have the respective values 4.0375 and -0.0045 stored in them. Perform the following in floating point arithmetic operations and find if there is any overflow/underflow.
 - a. fsub.s f5, f4, f3 [3]
 - b. fdiv.s f5, f3, f4 [3]
2. Why is a bias added to the actual exponent in the IEEE 754 floating-point representation, and how does this affect the encoding of both positive and negative exponents? [2]
3. How does optimized multiplication improve efficiency and performance compared to traditional long multiplication, especially in terms of speed and computational complexity? [2]
4. “mulh” instruction can be used to check for 64-bit overflow - Explain how with necessary examples. [2]
5. Convert the following code to RISC-V assembly language.

```
float a = fp_array [0];
float b = fp_array [1];
int c = 0;
if (a < b) {
    c = (int) a;
} else {
    c = (int) b;
}
```

Where fp_array is a floating point word array base address of which is stored in x13. [8]